Problem Statement

Land-use classification on EuroSAT RGB.

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EuroSAT (Helber et al., 2019) is a standard remote-sensing benchmark of 27,000 Sentinel-2 RGB patches (64×64 px) labeled with 10 land-use classes. Reliable land-use classification at scale is a prerequisite for downstream Earth-observation products such as urban-extent maps, deforestation alerts, and crop-type inventories.

Most published EuroSAT results are point comparisons: one architecture, one training recipe, one seed. This makes it difficult to attribute which modeling decision actually drove a reported gain — was it transfer learning, the optimizer, the regularizer, or simply more capacity?

We address three research questions:

- **RQ 1** How does test accuracy scale across a tiered hierarchy of feature extractors — raw pixels, handcrafted features, a shallow CNN, and a fine-tuned ResNet-18 — and at which tier do the marginal returns diminish?
- **RQ 2** Among six widely-used optimizers, which converges fastest and which generalizes best on the shallow CNN? Does the Wilson et al. (2017) claim that SGD with momentum dominates Adam hold on this dataset?
- **RQ 3** Among five regularization techniques (L2, Dropout, BatchNorm, Data Augmentation, Early Stopping), which most reduces the train/test gap on the shallow CNN, and how do they compose when stacked?

Proposed Solution

A controlled four-tier hierarchy on a single fixed split, with two ablations on the shallow CNN.

All models are trained and evaluated on an identical seed-42 stratified 70 / 15 / 15 split. The test set is touched exactly once per model.

TIER HIERARCHY

	FEATURE EXTRACTOR	CLASSIFIER	CAPACITY
T1	PCA on standardized raw pixels ($k \in \{50, 100, 200, 500\}$)	LR / LinearSVC / RBF-SVC (cuML)	*classical, raw pixels*
T2	HOG (1764-D) + per-channel color histogram (96-D) = 1860-D	LR / RBF-SVC (cuML)	*classical, handcrafted*
T3	Shallow CNN — 4 convolutional blocks (391K parameters)	end-to-end (cross-entropy)	*learned features*
T4	ResNet-18, ImageNet-pretrained, fully fine-tuned (3 seeds)	end-to-end (cross-entropy)	*deep transfer*

ABLATIONS ON TIER 3

EXPERIMENT	DESIGN	TOTAL RUNS
Experiment B — optimizer	6 optimizers × 3 learning rates × 3 seeds, 100 epochs each, no early stopping	54
Experiment C — regularization	9 regularization configurations × 3 seeds, Adam lr = 1e-3 fixed	27

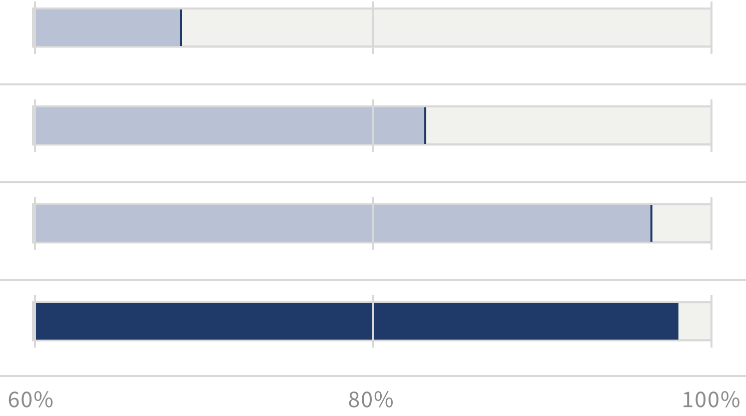
A GPU-resident `TensorDataset` reduces per-epoch time from **105 s to 1.06 s (≈ 100×)**, making the 54-run optimizer ablation tractable on a single A100.

Results & Conclusion

Test-set performance is monotonic across tiers; gains concentrate at the handcrafted-to-learned-features boundary.

TIER COMPARISON (BEST OF EACH TIER, SINGLE FIXED TEST SET)

	BEST MODEL	TEST ACCURACY (60% — 100%)	ACC.	MACRO-F1	Δ
T1	SVM-RBF on PCA(500)		0.6872	0.6826	—
T2	SVM-RBF on HOG + color histogram		0.8309	0.8271	+14.4 pp
T3	Shallow CNN + BatchNorm (3 seeds)		0.9657	0.9648	+13.5 pp
T4	**ResNet-18, fine-tuned (3 seeds)**		**0.9816**	**0.9812**	**+1.6 pp**



RQ 2 — OPTIMIZER

The top four optimizers (Adam, NAG, SGD+Momentum, RMSProp) are tied within 0.3 pp on final accuracy. Adam reaches 90% validation accuracy in 10 epochs versus 13–21 for the rest — it wins on convergence speed, not asymptotic accuracy. The Wilson et al. (2017) claim is not confirmed at this scale.

RQ 3 — REGULARIZATION

BatchNorm alone is the strongest single regularizer (+1.1 pp over baseline). Stacking all five techniques degrades accuracy by −3 pp relative to BatchNorm alone — over-regularization on a clean dataset. Per-class analysis identifies four crop classes (PermanentCrop, AnnualCrop, HerbaceousVegetation, Pasture) as the consistent bottleneck.

> **Conclusion.** *Representation learning (the T2 → T3 step) is where the accuracy gain lives; once a competent CNN with BatchNorm is in place, additional depth and optimizer choice yield diminishing returns.*